Fuel Processing Technology 85 (2004) 1121–1138

FUEL PROCESSING TECHNOLOGY

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Supercritical phase Fischer–Tropsch synthesis over cobalt catalyst

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Received 30 April 2003; received in revised form 1 September 2003; accepted 15 October 2003

Abstract

A supercritical phase Fischer–Tropsch reaction over a SiO_2 supported cobalt catalyst in a fixed bed reactor was studied in this paper. Four different supercritical fluids: *n*-pentane, *n*-hexane, *n*-heptane and *n*-decane were investigated. Similar CO conversion, hydrocarbon distribution, 1-olefin content of the products, etc. were obtained in *n*-pentane, *n*-hexane and *n*-heptane; however, 1-olefin content of the products in *n*-decane was obviously lower than that in *n*-pentane, *n*-hexane and *n*-heptane. Under the reaction conditions used in this study, CO conversion increased with increasing temperature and decreasing $W/F_{(CO+H_2)}$. The partial pressure of the *n*-hexane had little influence on the CO conversion under the conditions of that total pressure and syngas partial pressure kept constant. In general, the change of the distribution of the products was not obvious with the change of the operating conditions (time-on-stream, temperature, $W/F_{(CO+H_2)}$, etc.). The 1-olefin content of the products, however, increased remarkably with increasing partial pressure of *n*-hexane and decreasing temperature and $W/F_{(CO+H_2)}$. The reaction performances of the Fischer–Tropsch synthesis (FTS) in the supercritical phase fixed bed was also compared with those in the supercritical phase fluidized bed and the gas phase fixed bed. The results show that the 1-olefin content of the products in the supercritical phase fixed bed reactor was obviously higher than that in the gas phase fixed bed but slightly lower than that in the supercritical phase fluidized bed.

Keywords: Fischer–Tropsch synthesis; Supercritical phase; Cobalt catalyst; Fixed bed reactor

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0378-3820/$ - see front matter
doi:10.1016/j.fuproc.2003.10.015

1. Introduction

Fischer–Tropsch synthesis (FTS) has regained interest during the last decade as an option for the production of clean fuels and chemicals from either coal or natural gas due to changes in fossil energy reserves and environmental demands [1,2]. Because the FT reactions are highly exothermic, it is important to remove the heat of the reaction rapidly from the catalyst bed in order to avoid overheating of the catalyst, which would otherwise cause the deactivation of the catalyst due to sintering and fouling, and also an increase in the undesirable methane formation. Extensive research, therefore, has been focused on the improvement of heat removal of catalyst bed in the FT process.

The FTS has been studied or commercially operated in the fixed, slurry or fluidized bed reactors, and the features of the different types of reactors have been described in detail [1,3–5]. The FTS in the fixed bed is inevitably accompanied by the local superheating of the catalyst particles. The slurry bed, which is designed to overcome the defect of the fixed bed, is suffered from slow diffusion of reactant from the gas phase to the liquid phase as well as low space velocity. For the fluidized bed reactors, the efficiency of the heat transfer is higher than that in the fixed bed and the diffusion rate of reactant is faster than that in the slurry bed. A serious issue, however, is the possibility that heavy product deposits on the catalyst, which causes particles to agglomerate and thus leads to the defluidization of the bed. Therefore, further study on the improvement for the FT process, which has characteristics overcoming the above problems, is still necessary.

As a new field related to the supercritical fluid (SCF) technology, the use of SCFs as media for catalytic reactions has attracted increasing attention due to the unique properties of the supercritical fluids, such as high diffusion coefficient, high mass transfer efficiency and effective heat transfer capacity, etc. [6]. In recent studies of FTS, the researchers have introduced a supercritical phase into the conventional FT process [7–9]. It has been found that supercritical phase FTS (SFTS) showed some characteristics such as effective removal of reaction heat, quick diffusion of reactant gas and in situ extraction of high molecular weight hydrocarbons (wax), etc. Lang et al. [10] have investigated the SFTS on a precipitated iron catalyst (100 Fe/5Cu/4.2K/25SiO_2 on mass basis) in a fixed bed reactor. Propane was used as the SCF. The lumped hydrocarbon product distribution under supercritical conditions were similar to those obtained during reaction at the baseline conditions (gas phase conditions), however, higher selectivity of 1-olefin were obtained during SFTS. The authors considered that the increased 1-olefin selectivity during the SFTS is due to higher diffusivities and desorption rates of 1-olefin relative to normal FTS. Bochniak and Subramaniam [11], Subramaniam [12], Subramaniam and Snavely [13] have reported the SFTS results obtained from a Fe catalyst in a fixed bed reactor with near-critical *n*-hexane as SCF. They claimed that the catalyst effectiveness increased with pressure of the SCF due to the alleviation of pore-diffusion limitations. The results suggested to them that the pressure tuning of the density and transport properties of SCF offer a powerful tool to optimize catalyst activity and product selectivity during FTS. Huang and Roberts [14] have utilized a Co catalyst (15% Co–0.5% Pt/Al_2O_3) and a fixed bed reactor and reported that SFTS has a marked effect on the hydrocarbon product distribution with a shift to higher carbon number products owing to enhanced heat and mass transfer from catalyst surface.

In the previous studies by our group [15–20], we investigated the reaction performances of SFTS from the viewpoints of extraction capability and mass diffusion efficiency. We also compared the results obtained from three different reaction phases: gas, liquid and supercritical phase. *n*-Hexane was employed as SCF and three different catalysts: Co, Fe and Ru catalysts were used, respectively. The results suggested that, although the rate of the reaction and the diffusion of the reactants in the SFTS were slightly lower than those in the gas-phase reaction, the removal of reaction heat and waxy products from the catalyst surface was much more effective than in the gas-phase reaction. The olefin content in the hydrocarbon products was much higher in the SFTS than in the liquid and gas phase reactions. As a part of NEDO-ATL (Asphalt to Liquid) Project (Japan), this work mainly focuses on the development of the SFTS process. So the effects of supercritical fluids, time-on-stream, reaction temperature, partial pressure of supercritical solvent, $W/F_{(CO+H_2)}$ (W/F) on the reaction behaviors of the SFTS in a fixed bed reactor were systematically investigated. The results were also compared with those obtained in the supercritical phase fluidized bed and the gas-phase fixed bed reactors.

2. Experimental

2.1. Catalyst preparation

The catalyst used was a SiO_2 supported cobalt catalyst, which was prepared by impregnating a commercially available Q-15 SiO_2 gel (Fuji Silysia Chemical, Japan) with cobalt nitrate from its aqueous solution by incipient wetness impregnation method. Composition of the catalyst was Co: 20 and SiO_2: 80 by weight. The catalyst precursor was dried at 120 °C overnight in an air oven and then calcined at 200 °C for 2 h in air atmosphere to form supported metal oxides.

2.2. Reaction apparatus and procedure

The reaction apparatus flow sheet for the SFTS is shown in Fig. 1. This flow sheet consists of two parts: the fixed bed reactor system and the fluidized bed reactor system. The flow sheet of the fixed bed was similar to that of a conventional, pressurized, fixed bed reactor system, except that a preheating vaporizer and an ice-cooled high-pressure trap were set upstream and downstream of the reactor, respectively. SCF, liquid hydrocarbons and wax were exhausted from the ice-cooled trap continuously, and uncondensed products with unreacted syngas were expanded and then introduced to on-line gas chromatograph. The flow sheet of the fluidized bed was similar to that of the fixed bed. The only difference was that the syngas and supercritical solvent were introduced into the fluidized bed from the bottom of the reactor.

The catalyst was packed into the reactor and reduced in situ in hydrogen flow at 400 °C for 3 h before the reaction. After the reduction, hydrogen and supercritical solvent at certain ratio were directed to the reaction system and the temperature and pressure of the reaction system were raised simultaneously. Once the temperature and pressure reached desired values, hydrogen was replaced by syngas and the reaction time was counted. Each

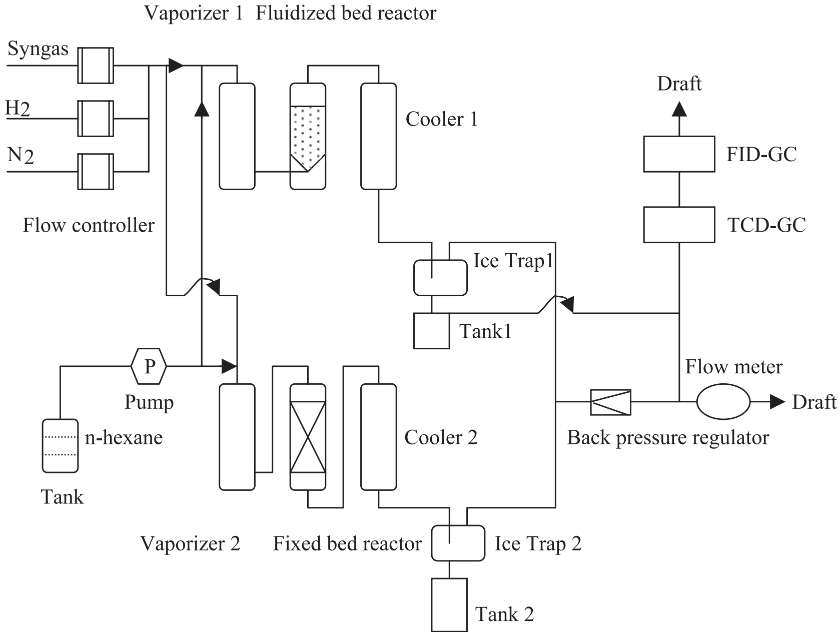

Fig. 1. Flow scheme of apparatus for supercritical Fischer–Tropsch synthesis.

reaction was conducted for 6 h. *n*-Pentane (n-C_5), *n*-hexane (n-C_6), *n*-heptane (n-C_7) and *n*-decane (n-C_{10}) were employed as the supercritical solvents, respectively. The amount of the catalyst loaded and the mean diameter of the catalyst were 1 g and 530 μm for the fixed bed and 2.5 g and 285 μm for the fluidized bed. A typical reaction conditions were $T=240$ °C, $P_{total}=4.5$ MPa, $P_{(CO+H_2)}=1.0$ MPa, $P_{solvent}=3.5$ MPa and W/F = 5 g cat. h/mol. Argon was used as the internal standard in the feed gas. For the comparison, nitrogen was used as balance gas in some runs.

Gaseous compounds were analyzed on-line by two-coupled gas chromatographs. CO, CO_2 and CH_4 were analyzed by using an activated charcoal column with a thermal conductivity detector (TCD). Light hydrocarbons (C_1–C_6) were analyzed by using an Al_2O_3–KCl capillary column with a flame ionization detector (FID). A capillary column (DB-2881) with a FID was used for the analysis of the liquid compounds. The *n*-hexadecane was used as the internal standard for liquid products. The chain growth probability (α) of the products was defined by the Anderson–Schulz–Flory plot, in which the carbon number extended from C_4 to C_{20} [21].

3. Results and discussion

3.1. Dynamic behavior of the SFTS in the fixed bed reactor

Fig. 2 shows CO conversion, CH_4 and CO_2 selectivities as a function of time-on-stream at 240 °C, 4.5 MPa of total pressure and 3.5 MPa of *n*-hexane partial pressure.

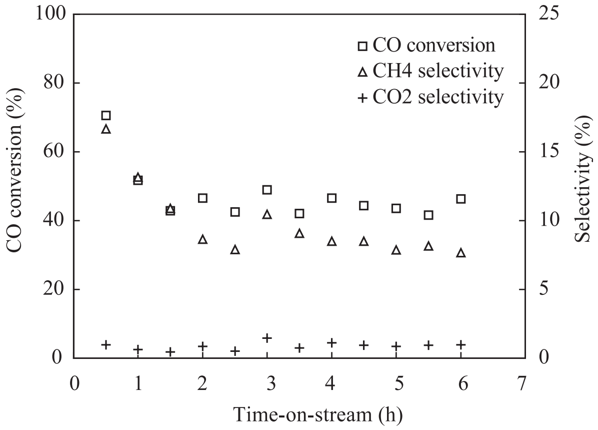

Fig. 2. Effect of time-on-stream on CO conversion, CH_4 and CO_2 selectivities. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

CO conversion represented as open square decreased from about 71% at 0.5 h to a steady state value of about 45% in about 3 h. This decrease is accompanied by a decrease of CH_4 selectivity represented as open triangle, which decreased from about 17% to 8% as time-on-stream increased from 0.5 to 2.0 h, and then kept at about 8%. The value of CO_2 selectivity, however, was always about 0.9% independent of time-on-stream. In fact, for all the experiments (except a run for comparison presented in the below) presented in this paper, similar trends for CO conversion, CH_4 and CO_2 selectivities with time-on-stream were observed. Decreasing CO conversion with increasing reaction time is reasonable since all of the active sites of the catalyst are active at the begin of reaction and then some of active sites are poisoned or covered by other species with time-on-stream which results in the gradual decrease of the catalyst activity at the initial stage. Higher CH_4 selectivity at the initial stage may be caused by high H_2 concentration on the surface of the catalyst at the initial stage. Since hydrogen was employed to pressurize the reaction system, certain amount of H_2 has been adsorbed on the surface of the catalyst before the reaction started. Hence, the H_2/CO ratio of the catalyst surface was higher than that of feed gas at the initial stage, which resulted in high CH_4 formation. The phenomenon is consistent with the result that higher H_2 partial pressure favors the formation of methane [1,22]. Furthermore, as a comparison, another run was carried out under the same reaction conditions except that nitrogen was used to pressurize the reaction system before the reaction, the result of which is shown in Fig. 3. It can be seen that CH_4 selectivity always kept at about 8%, irrespective of the change of the reaction time. This again supports the above explanation.

Fig. 4 presents the effect of the time-on-stream on the selectivity of hydrocarbons at 240 °C and 4.5 MPa. Except for methane, the maximum selectivities of hydrocarbons were at C_5, C_8 and C_8, corresponding to the time-on-stream of 1, 4 and 6 h. In general, the selectivity of hydrocarbons had little change with an increase in reaction time. The chain growth probability was almost constant ($\alpha = 0.85$), irrespective of the time-on-stream. Further result indicates that the time-on-stream had effect on the 1-olefin content of the

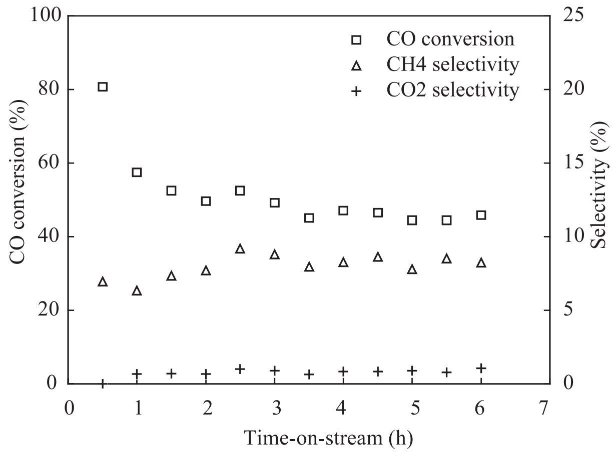

Fig. 3. Effect of time-on-stream on CO conversion, CH_4 and CO_2 selectivities under the condition of that hydrogen was replaced by nitrogen to pressurize the system before the reaction. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

products, which is shown in Fig. 5. The 1-olefin content of the products in time of 4 h was slightly higher than that of 1 h but close to that of 6 h. Especially for light and middle fractions (C_2–C_{15}), the change of the 1-olefin content with time-on-stream was relatively obvious. For all the experiments, except C_2, the 1-olefin content of the hydrocarbons decreased with increased carbon number. The lower 1-olefin content in time of 1 h might be due to the higher H_2 concentration on the catalyst surface at the initial stage that likely leads to the termination of the surface species to the paraffins.

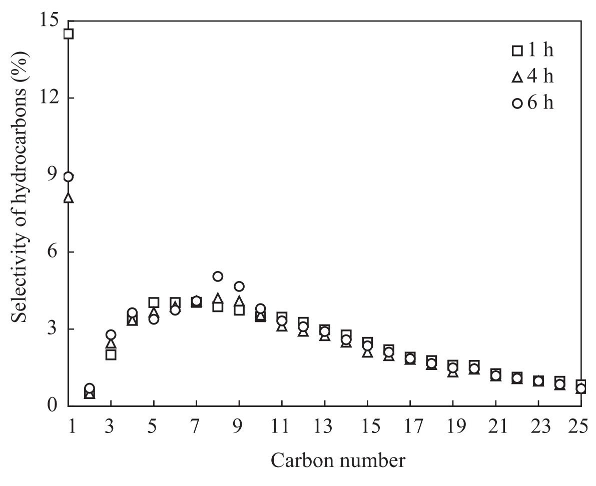

Fig. 4. Effect of time-on-stream on the selectivity of the products. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

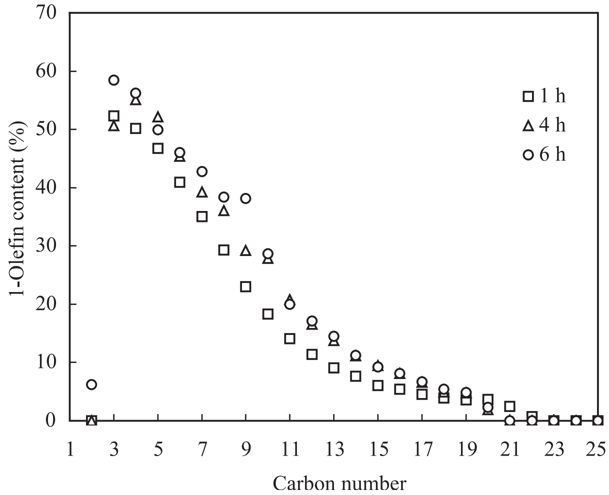

Fig. 5. Effect of time-on-stream on the 1-olefin content of the products. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

3.2. Effect of solvents on the SFTS in the fixed bed reactor

The critical constants of the solvents and the reaction behavior of SFTS in the different solvents were listed in the Table 1. Under the reaction conditions used in this study ($T = 240$ °C, $P_{solvent} = 3.5$ MPa), *n*-pentane and *n*-hexane were at supercritical conditions and *n*-heptane was at near-critical conditions; however, *n*-decane was far from the critical conditions due to the higher critical temperature (344.7 °C) than the reaction temperature (240 °C). It should be noted that CO conversion, CH_4 and CO_2 selectivities were average values of the data obtained at the steady state for about three hours (below is same). It is surprised that all of CO conversion, CH_4, CO_2 selectivities and α value changed little by changing of the solvent. The effects of the solvents on the selectivity of hydrocarbons and 1-olefin content of the products were shown in Figs. 6 and 7, respectively. Relatively, the products carbon distribution in *n*-pentane, *n*-hexane and *n*-heptane were similar and higher C_4–C_8 selectivity was obtained in *n*-decane media. The 1-olefin content of the hydrocarbons in *n*-pentane and *n*-hexane were similar and slightly lower than that in *n*-

Table 1
The critical constants of solvents and the reaction performance of FTS in different solvents

Solvent	Critical constants			Reaction performance		
	T_c (°C)	P_c (MPa)	CO conv. (%)	CH_4 sel. (%)	CO_2 sel. (%)	α
n-C_5H_{12}	196.5	3.37	43.9	9.3	1.1	0.84
n-C_6H_{14}	234.3	2.97	45.4	8.3	0.8	0.83
n-C_7H_{16}	266.9	2.74	46.6	10.6	0.8	0.84
n-$C_{10}H_{22}$	344.7	2.11	45.9	9.8	0.7	0.82

Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{solvent} = 3.5$ MPa and W/F = 5 g cat. h/mol.

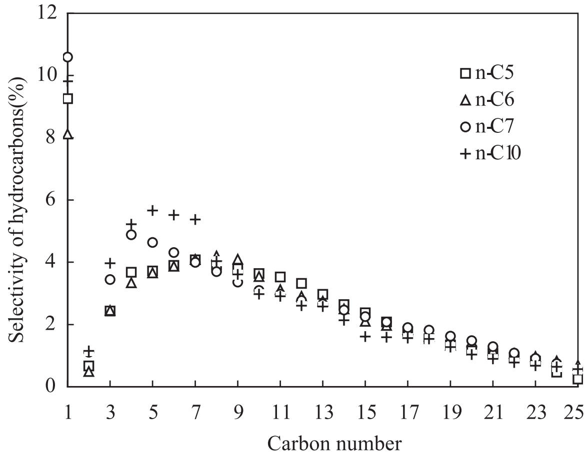

Fig. 6. Effect of solvent on products distribution. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{solvent} = 3.5$ MPa and W/F = 5 g h/mol.

heptane, but obviously higher than that in *n*-decane. These results indicate that the extraction capacity of *n*-decane to 1-olefins, primary products of the FTS, is lower than that of *n*-pentane, *n*-hexane and *n*-heptane. That *n*-decane was at far from critical condition was probably one of reasons for the lower 1-olefin content of the products in *n*-decane media. The result that *n*-pentane, *n*-hexane and *n*-heptane have similar effects on the SFTS suggested that the recycle of the SCF contained lighter fractions (such as C_4-C_8) produced in the FT process is possible for reducing the amount of supercritical solvent. This is important for the possible application of supercritical phase reaction to practical FTS.

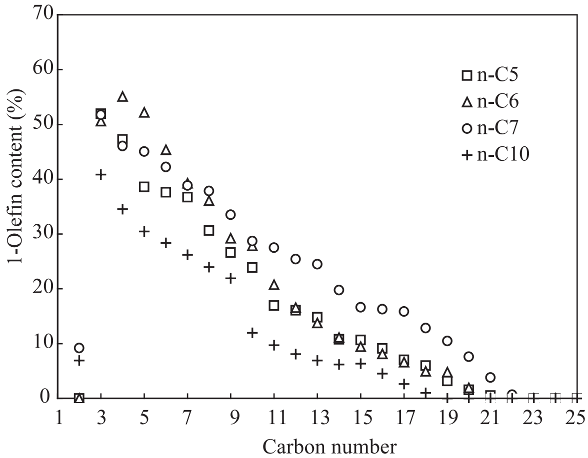

Fig. 7. Effect of solvent on 1-olefin content of hydrocarbons. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{solvent} = 3.5$ MPa and W/F = 5 g h/mol.

3.3. Effect of reaction temperature on the SFTS in the fixed bed reactor

Fig. 8 shows CO conversion, CH_4 and CO_2 selectivities as a function of reaction temperature. CO conversion increased with increasing reaction temperature and CH_4 and CO_2 selectivities as well. It should be noted that the changes of CO conversion, CH_4 and CO_2 selectivities with reaction temperature were not linear. Compared with CO conversion, CH_4 and CO_2 selectivities at 220 °C, the corresponding values at 230 °C only increased 6.2%, 0.5% and 0.1%, respectively. When the reaction temperature increased from 240 to 250 °C, however, the corresponding values increased as high as 20.8%, 5.6% and 3.7%, respectively. These results are reasonable because higher temperatures favor not only the conversion of CO but also the formation of CH_4. Carbon dioxide is mostly produced by the WGS reaction from the primary by-product, water. The higher CO conversion, the more water formation. This enhances the proceeding of the WGS reaction.

The influence of reaction temperature on the selectivity and the 1-olefin content of the products are shown in Figs. 9 and 10, respectively. As the operating temperature was increased, the products selectivity slightly shifted to lighter molecular mass compounds. Relatively, the distribution of hydrocarbons became flat with increasing reaction temperature. The value of the chain growth probability decreased from 0.86 to 0.82 when the temperature increased from 220 to 250 °C. It can be seen from Fig. 8 that the 1-olefin content remarkably decreased with increasing temperature. The average 1-olefin content of the products at 220 °C was about triple that at 250 °C. Since cobalt is a very active hydrogenating catalyst, the primary products, 1-olefins, in general are more hydrogenated with increasing temperature. In addition, higher temperature also favors the secondary reactions (isomerization, hydrogenation and readsorption) of the 1-olefin. These shifts are in line with thermodynamic expectations and the relative stability of the products [1].

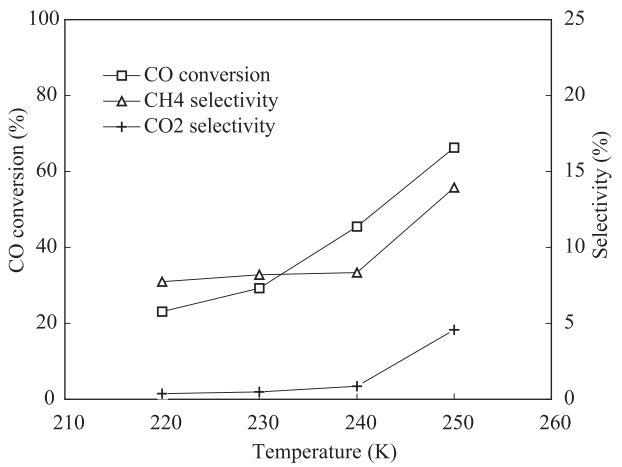

Fig. 8. Effect of reaction temperature on CO conversion, CH_4 and CO_2 selectivities. Reaction conditions: $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

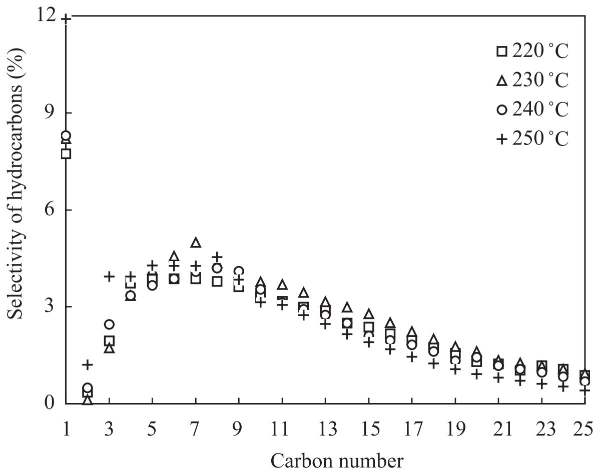

Fig. 9. Effect of temperature on the selectivity of the products. Reaction conditions: $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

3.4. Effect of W/F on the SFTS in the fixed bed reactor

CO conversion, CH_4 and CO_2 selectivities at different W/F are shown in Fig. 11. CO conversion increased from 30.3% to 86.5% as W/F increased from 2.5 to 10 g cat. h/mol. CH_4 selectivity rose slowly with increasing W/F. W/F had little influence on CO_2 selectivity at lower values (in our case is at the range of 2.5–5.0 g cat. h/mol). However, when the value of W/F increased from 7.5 to 10.0 g cat. h/mol, the corresponding CO_2 selectivity increased rapidly from 1.6% to 4.1%. The residence time of synthesis gas in the

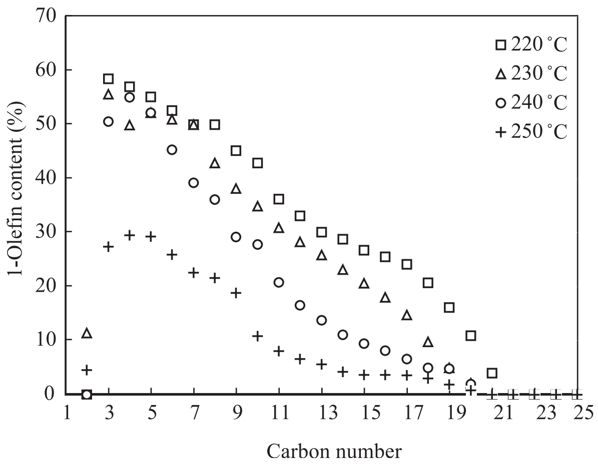

Fig. 10. Effect of temperature on the 1-olefin content of the products. Reaction conditions: $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

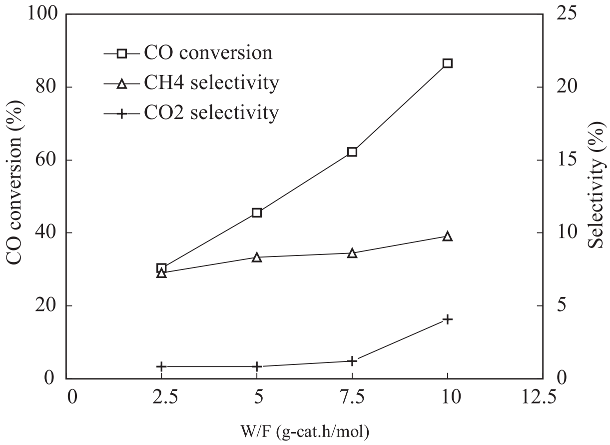

Fig. 11. Effect of W/F on CO conversion, CH_4 and CO_2 selectivities. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa and $P_{hexane} = 3.5$ MPa.

catalyst bed becomes longer with the increase of W/F, which favors more CO was hydrogenated. So it is reasonable that CO conversion increases rapidly with increasing W/F. One of the reasons for the higher CO_2 selectivity at higher W/F might be that the amount of the by-product, water, increases with increased CO conversion, which promotes the formation of CO_2. Another probable reason is that the removal rate of the water from the catalyst bed at higher W/F is slower than that at lower W/F.

Figs. 12 and 13 present the selectivity and 1-olefin content of the products at different W/F. The distribution of hydrocarbons at W/F of 5 g cat. h/mol is boarder than that at W/F

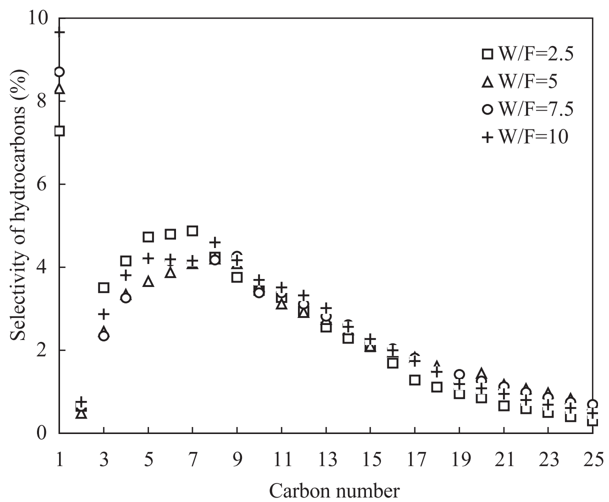

Fig. 12. The selectivity of the products at different W/F. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa and $P_{hexane} = 3.5$ MPa.

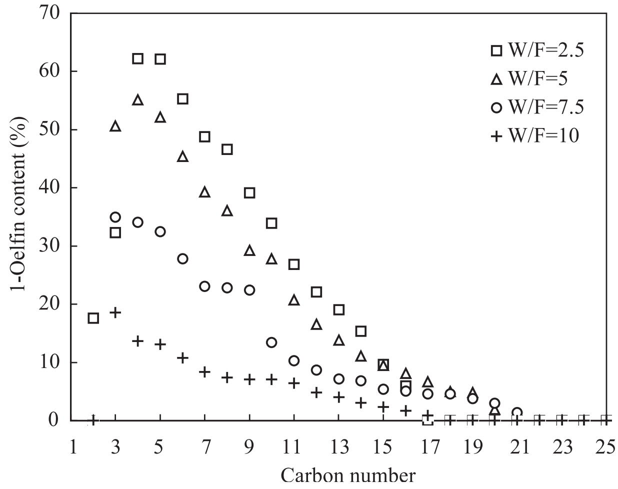

Fig. 13. The 1-olefin content of the products at different W/F. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa and $P_{hexane} = 3.5$ MPa.

of 2.5 g cat. h/mol but very similar to that at W/F of 7.5 and 10 g cat. h/mol. This result seems to indicate that in some degree higher W/F enhances the production of heavier fractions. The calculation of α value is also consistent with the above result. The value of α increased from 0.80 to 0.84 when reaction W/F increased from 2.5 to 10 g cat. h/mol. With the decrease of W/F, the flow rate of *n*-hexane increased and the products was extracted or washed from the surface of catalyst more quickly, which may suppress the growth of carbon chains of the products. 1-Olefin content of the products decreased drastically with increasing W/F. These results should be mainly attributed to the increase in residence time of both reactant and products with increasing W/F. The longer residence time is favorable to the secondary reactions of the products, such as hydrogenation and oligomerization.

3.5. Effect of partial pressure of n-hexane on the SFTS in the fixed bed reactor

The effect of partial pressure of the supercritical solvent, *n*-hexane, on CO conversion, CH_4 and CO_2 selectivities is shown in Table 2. In this study, the overall pressure and the partial pressure of the synthesis gas were kept constant and only the partial pressure of the supercritical fluid was changed, while a balancing inert gas (nitrogen) was fed to maintain constant space velocity. It can be seen from Table 2 that CO conversion, CH_4 and CO_2

Table 2
CO conversion, CH_4 and CO_2 selectivities at different partial pressures of *n*-hexane

P_{hexane} (MPa)	P_{total} (MPa)	P_{syngas} (MPa)	P_{N_2} (MPa)	CO conv. (%)	CH_4 sel. (%)	CO_2 sel. (%)
3.5	4.5	1.0	0	45.5	8.3	0.9
2.5	4.5	1.0	1.0	43.3	7.1	0.8
1.5	4.5	1.0	2.0	44.4	9.7	1.0

Reaction conditions: $T = 240$ °C and W/F = 5 g cat. h/mol.

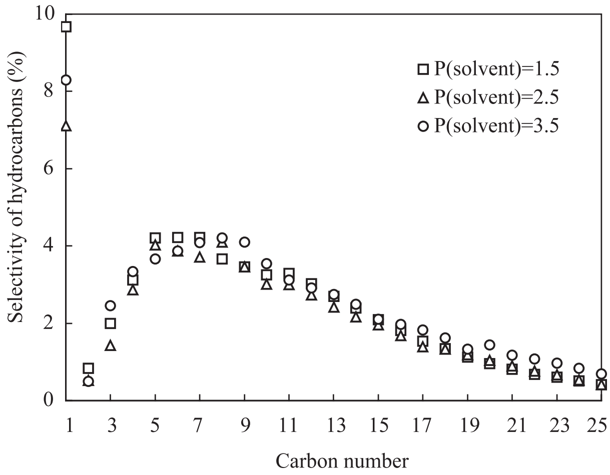

Fig. 14. The selectivity of the products at different partial pressure of n-hexane. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa and W/F = 5 g cat. h/mol.

selectivities had little changes irrespective of the partial pressure of n-hexane. The partial pressure of n-hexane also had little influence on the hydrocarbon distribution of products, as shown in Fig. 14. As we all know that the introduction of the supercritical fluid is mainly for the improvement of the heat transfer of the catalyst bed, in-situ extraction of the heavier products, etc., but it does not change the mechanisms of the FT reactions. This may be a probable reason but need to be identified by the further research. The effect of n-hexane partial pressure on 1-olefin content of the products is shown in Fig. 15. In general, 1-olefin content increased with increasing n-hexane partial pressure. This is expected

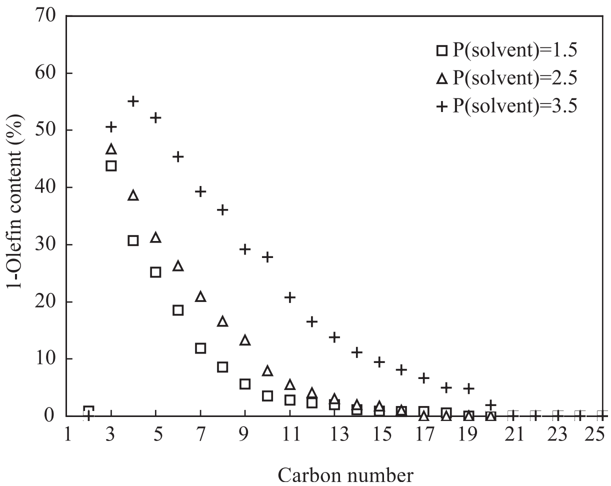

Fig. 15. The 1-olefin content of the products at different partial pressure of n-hexane. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa and W/F = 5 g cat. h/mol.

because the extraction ability is improved with the increase of *n*-hexane partial pressure. This results in an increase in pore diffusion efficiency and an increase in the solubility of heavy products from the inner pores of the catalyst, which suppresses the secondary reactions of 1-olefin. It also should be noted that 1-olefin content at 2.5 MPa of *n*-hexane partial pressure was slightly higher than that at 1.5 MPa of *n*-hexane partial pressure but obviously lower than that at 3.5 MPa of *n*-hexane partial pressure. We know that the critical pressure of *n*-hexane (Table 1) is 2.97 MPa. Therefore, when *n*-hexane partial pressure (such as 2.5 and 1.5 MPa in the case) is lower than the critical pressure of *n*-hexane the system in fact is not at supercritical conditions. These results indicate that in-situ extraction capacity of the solvent at the supercritical conditions is higher than that at non-supercritical conditions.

3.6. Comparison of reaction behavior of FTS in the different reaction phase

The comparison of SFTS in three different reaction phases: the supercritical phase fixed bed, the supercritical phase fluidized bed and the gas phase fixed bed were studied in this work. To compare the three reaction phases in the same conditions, nitrogen was used as the balance gas for the gas phase reaction. Fig. 16 shows the temperature distribution of the catalyst bed in the three reaction phases. The maximum temperature rise in the catalyst bed was in the following order: supercritical phase fluidized bed (11 °C) < the supercritical phase fluidized bed (15 °C) < the gas phase fixed bed (21 °C). It is expected that the heat transfer of the catalyst bed in the SCF is more effective than that in the gas phase.

Table 3 shows some of the reaction performances of FTS in the three reaction systems. It can be seen that CO conversion was in the following order: supercritical phase fixed bed (45.4%) < supercritical phase fluidized bed (47.3%) < gas phase fixed bed (53.4%). This fact seems to indicate that the diffusion of synthesis gas in the supercritical phase fluidized

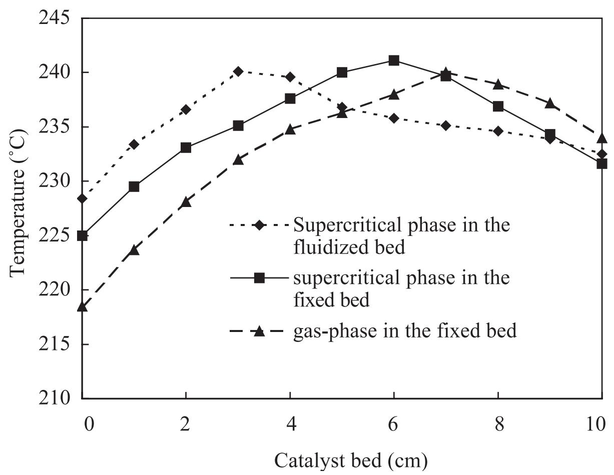

Fig. 16. Temperature profiles of the catalyst bed in different reaction phases. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

Table 3
The comparison of the FTS in the different reaction systems

Reaction phase	Supercritical phase fluidized bed	Supercritical phase fixed bed	Gas phase fixed bed
CO conv. (%)	47.3	45.4	53.4
CH_4 sel. (%)	9.1	8.3	11.1
CO_2 sel. (%)	0.9	0.8	1.6
Chain growth probability	0.82	0.85	0.84
Carbon balance	82.3	85.1	87.5

Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

bed was faster than that in the supercritical phase fixed bed but slower than that in the gas phase fixed bed. It should also be noted that the methane and CO_2 selectivities in the gas phase is higher that in the supercritical phase. In general, high temperature is favorable for the formation of methane. Therefore, the low methane selectivities both in the supercritical phase fluidized bed and in the supercritical phase fixed bed should be attributed to the more effective removal of reaction heat from the catalyst surface than in the gas phase fixed bed. In our prior study, it has been found that *n*-hexane can also extract water in the supercritical phase reaction [20]. Thus, the lower carbon dioxide selectivity in the supercritical phase might be due to quickly removal of water from catalyst bed, which suppresses the occurrence of the WGS reaction.

The selectivity of the products in the different reaction systems was shown in Fig. 17. The highest selectivities of hydrocarbons were at C_5 (4.8%) for the fluidized bed reactor, C_8 (4.2%) for the supercritical phase fixed bed and C_{10} (4.3%) for the gas phase fixed bed. Relatively, the hydrocarbons produced in the fluidized bed were slightly richer in the light fractions than that in the fixed beds. It can also be seen from Table 3 that the chain growth probability in the fluidized bed (0.82) was lower than that in the fixed bed (0.85 for the

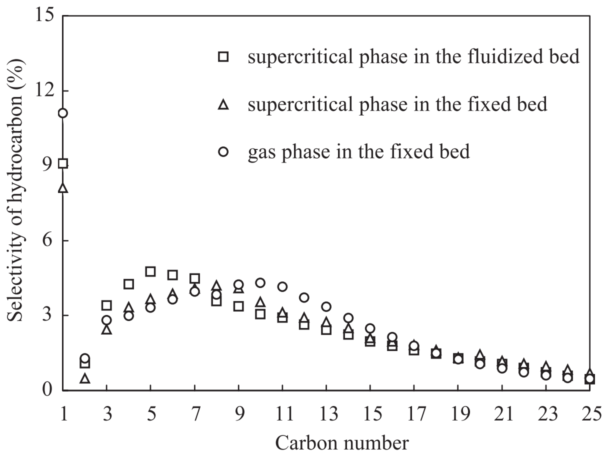

Fig. 17. The selectivity of the products in different reaction phases. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

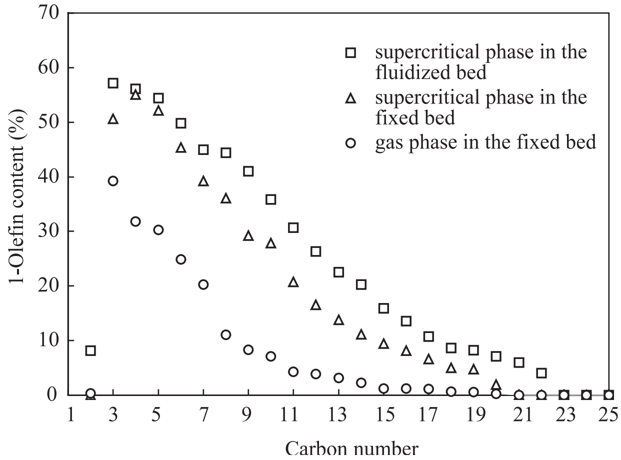

Fig. 18. The 1-olefin content of the products in different reaction phases. Reaction conditions: $T = 240$ °C, $P_{total} = 4.5$ MPa, $P_{(CO+H_2)} = 1.0$ MPa, $P_{hexane} = 3.5$ MPa and W/F = 5 g cat. h/mol.

supercritical phase and 0.84 for the gas phase). One of reasons for these results might be caused by the smaller particle size of the catalyst used in the fluidized bed reactor. Further study, however, is necessary for the really reasons. Fig. 18 presents the effect of reaction systems on the 1-olefin content of the products. The 1-olefin content of the products in the supercritical phase fixed bed was obviously higher than that in the gas phase fixed bed but slightly lower than that in the supercritical phase fluidized bed. That 1-olefin content of the products in the SFTS is higher than that in the gas phase FTS is consistent with the result reported by other researchers [10,14]. This phenomenon suggests that the 1-olefins, the primary products of the FT reaction, can be effectively extracted and transported by the supercritical fluid out of the catalyst particles both in the fixed bed and in the fluidized bed before they were readsorbed and hydrogenated to the paraffins.

4. Conclusions

The SFTS over a silica supported cobalt catalyst in a fixed bed reactor was investigated in this paper. The effects of SCFs, time-on-stream, temperature, W/F and partial pressure of *n*-hexane on the reaction behavior were mainly studied.

Four different SCFs, *n*-pentane, *n*-hexane, *n*-heptane and *n*-decane, were investigated. Similar CO conversion, hydrocarbon distribution, 1-olefin content, etc. were obtained in *n*-pentane, *n*-hexane and *n*-heptane; however, 1-olefin content of the products in *n*-decane was obviously lower than that of in *n*-pentane, *n*-hexane and *n*-heptane. That *n*-decane was far from critical condition under the reaction temperature was probably one of reasons for the lower 1-olefin content of the products in *n*-decane fluid. The result that *n*-pentane, *n*-hexane and *n*-heptane have similar effects on the SFTS suggested that the recycle of the SCF contained lighter fractions (such as C_4–C_8) produced in the FT process is possible for reducing the amount of supercritical solvent.

Under the reaction conditions used in this work, CO conversion decreased with time on stream at the initial stage and then remained almost constant at the steady state as well as CH_4 selectivity. CO_2 selectivity, however, had little change, irrespective of time on stream. At the steady state, CO conversion increased with increasing temperature and decreasing W/F. The partial pressure of the *n*-hexane had little influence on the CO conversion when total pressure and syngas partial pressure kept constant. CH_4 and CO_2 selectivity increased with increasing temperature and W/F. In general, the change of the distribution of the products was not obvious with the change of the operating conditions (time-on-stream, temperature, W/F, etc).The 1-olefin content of the products, however, increased remarkably with increasing partial pressure of *n*-hexane and decreasing temperature and W/F, respectively.

The reaction performance of the FTS in the supercritical phase fixed bed was also compared with those in the supercritical phase fluidized bed and the gas phase fixed bed. Low methane selectivity was obtained in the supercritical phase due to the effective removal of the heat from the catalyst bed by the SCF. The 1-olefin content of the products in the supercritical phase fixed bed was obviously higher than that in the gas phase fixed bed but slightly lower than that in the supercritical phase fluidized bed This result suggests that the primary 1-olefins can be effectively extracted and transported by the SCF out of the catalyst particles both in the fixed bed and in the fluidized bed before they were readsorbed and hydrogenated to the paraffins.

Acknowledgements

This work was entrusted from Nippon Oil as a part of NEDO-ATL Project (Japan).

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